

Why scientists love games consoles



PS3 console: 'You can build your own supercomputer using PS3s'

By Roger Highfield, Science Editor

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Leading scientists are turning to the extraordinary power of games consoles to do their sums and simulate everything from colliding black holes to the effects of drugs.

Supercomputer 'virtual human' to help fight disease

Doctors use Nintendo Wii in therapy treatment

PlayStation3 Gravity Grid

Reprogram a PlayStation and it will perform feats that would be unthinkable on an ordinary PC because the kinds of calculations required to produce the realistic graphics now seen in sophisticated video games are similar to those used by chemists and physicists as they simulate the interactions between particles ranging from the molecular to the astronomical.

Such simulations are usually carried out on a supercomputer, but time on these machines is expensive and in short supply. By comparison, games consoles are cheap and easily available, says New Scientist.

"There is no doubt that the entertainment industry is helping to drive the direction of high performance computational science - exploiting the power available to the masses will lead to many research breakthroughs in the future," comments Prof Peter Coveney of University College London, who uses supercomputing in chemistry.

Prof Gaurav Khanna at the University of Massachusetts has used an array of 16 PS3s to calculate what will happen when two black holes merge.

According to Prof Khanna, the PS3 has unique features that make it suitable for scientific computations,

namely, the Cell processor dubbed a "supercomputer-on-a-chip." And it runs on Linux, "so it does not limit what you can do."

"A single high-precision simulation can sometimes cost more than 5,000 hours on the TeraGrid supercomputers. For the same cost, you can build your own supercomputer using PS3s. It works just as well, has no long wait times and can be used over and over again, indefinitely," Prof Khanna says.

And Todd Martínez has persuaded the supercomputing centre at the University of Illinois, Urbana-Champaign, to buy eight computers each driven by two of the specialised chips that are at the heart of Sony's PlayStation 3 console.

Together with his student Benjamin Levine he is using them to simulate the interactions between the electrons in atoms, as part of work to see how proteins in the body dovetail with drug molecules.

He was inspired while browsing through his son's games console's technical specification "I noticed that the architecture looked a lot like high performance supercomputers I had seen before," he says. "That's when I thought about getting one for myself."

An effort to interconnect tens of thousands of PS3s is under way with Folding@Home, an effort based at Stanford University to study the way proteins fold, which plays a key role in Alzheimer's, Huntington's Disease and Parkinson's disease.

With about 50,000 such machines, the organisers of this huge distributed computing effort hope to achieve performance on the petaflop scale.

The Wii, made by Nintendo, has a motion tracking remote control unit that is cheaper than a comparable device built from scratch. The device recently emerged as a tool to help surgeons to improve their technique.

Meanwhile, neurologist Thomas Davis at the Vanderbilt Medical Centre in Nashville, Tennessee, is using it to measure movement deficiencies in Parkinson's patients to assess how well a patient can move when they take part in drug trials.

How we moderate

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